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Artificial Intelligence–Assisted Data Extraction With a Large Language Model: A Study Within Reviews

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Background: Data extraction is a critical but error-prone and labor-intensive task in evidence synthesis. Unlike other artificial intelligence (AI) technologies, large language models (LLMs) do not require labeled training data for data extraction.

Objective: To compare an AI-assisted versus a traditional, human-only data extraction process.

Design: Study within reviews (SWAR) using a prospective, parallel-group comparison with blinded data adjudicators.

Setting: Workflow validation within 6 ongoing systematic reviews of interventions under real-world conditions.

Intervention: Initial data extraction using an LLM (Claude, versions 2.1, 3.0 Opus, and 3.5 Sonnet) verified by a human reviewer.

Measurements: Concordance, time on task, accuracy, sensitivity, positive predictive value, and error analysis.

Results: The 6 systematic reviews in the SWAR yielded 9341 data elements from 63 studies. Concordance between the 2 methods was 77.2% (95% CI, 76.3% to 78.0%). Compared with the reference standard, the AI-assisted approach had an accuracy of 91.0% (CI, 90.4% to 91.6%) and the human-only approach an accuracy of 89.0% (CI, 88.3% to 89.6%). Sensitivities were 89.4% (CI, 88.6% to 90.1%) and 86.5% (CI, 85.7% to 87.3%), respectively, with positive predictive values of 99.2% (CI, 99.0% to 99.4%) and 98.9% (CI, 98.6% to 99.1%). Incorrect data were extracted in 9.0% (CI, 8.4% to 9.6%) of AI-assisted cases and 11.0% (CI, 10.4% to 11.7%) of human-only cases, with corresponding proportions of major errors of 2.5% (CI, 2.2% to 2.8%) versus 2.7% (CI, 2.4% to 3.1%). Missed data items were the most frequent error type in both approaches. The AI-assisted method reduced data extraction time by a median of 41 minutes per study.

Limitations: Assessing concordance and classifying errors required subjective judgment. Consistently tracking time on task was challenging.

Conclusion: Data extraction assisted by AI may offer a viable, more efficient alternative to human-only methods.

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Among all steps involved in systematic reviews, data extraction (that is, the process of extracting data from primary studies into standardized tables) is one of the most critical, labor-intensive, and error-prone tasks (1). Although single-investigator data extraction with secondary verification requires, on average, 107 minutes per study (2), methodological evaluations identified extraction errors in up to 67% of meta-analyses (3). Common errors include data omissions, statistical misclassifications, misinterpretations of ambiguous primary studies, and data entry mistakes.

The use of artificial intelligence (AI) may be able to increase efficiency and correctness of the data extraction process (4). Particularly, the introduction of large language models (LLMs), such as generative pretrained transformers (GPTs) (5), has opened new possibilities for semiautomated data extraction, which combines machine-learning capabilities with human oversight. Unlike earlier natural language processing technologies used for data extraction, pretrained LLMs do not require task-specific training data, making them more accessible to users without technical expertise.

Initial evaluations of LLMs for data extraction demonstrated variable accuracy, ranging from 72% to 100%, compared with human reference standards (6–11). However, these studies' reliance on controlled experimental conditions and use of preexisting review data sets as benchmarks limit their generalizability to real-world applications. Furthermore, they assessed fully automated approaches without human involvement and evaluated the LLMs outside the actual workflow of an evidence synthesis. Although LLMs

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can potentially save time in data extraction, defining prompts (that is, the specific inputs provided to a generative AI system to elicit a response) and interacting with the models add additional tasks to the review process. Therefore, validating their use within real-world workflows, specifically measuring both accuracy and time spent, is essential.

Our study aimed to examine a prospective workflow for an AI-assisted data extraction approach in which an LLM replaced a human investigator during the initial data extraction. We selected a semiautomated approach because a fully automated method without human oversight is unlikely to be adopted for evidence synthesis in the near future.

METHODS

We registered the protocol of this study with the Northern Ireland Network for Trials Methodology Research (SWAR28; www.qub.ac.uk/sites/TheNorthernIrelandNetworkforTrialsMethodologyResearch/FileStore/SWARFileStore/SWAR28%20Gerald%20Gartlehner%20(2023%20FEB%2011%202102).pdf). We follow reporting guidance of the TRIPOD-LLM (Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis extension for Large Language Models) statement (12).

Research Questions

Our study was guided by the following 2 research questions. First, what is the concordance between human-only and AI-assisted, semiautomated data extraction processes when used in real-world systematic review workflows, and how does the time on task compare between these 2 methods? Second, what is the accuracy of the AI-assisted data extraction process, and how do the types of errors compare with those of the human-only data extraction?

Study Design

This study used a study within reviews (SWAR) (13) design incorporating 6 ongoing systematic reviews (14–19) conducted by the Agency for Healthcare Research and Quality's Evidence-based Practice Center Program. The systematic reviews addressed diverse topics and included both randomized and nonrandomized studies. Except for 1, each systematic review was enrolled in the SWAR after its review team had completed the literature screening phase. If a review included more than 20 studies, an investigator not involved in the review randomly selected 20 studies for the SWAR by generating a list of random numbers and assigning a single number to each study. The 20 studies with the lowest numbers were then included for AI-assisted data extraction. Because of time constraints and an approaching SWAR deadline, 1 review included only 7 randomly selected studies.

For each participating systematic review, we did a prospective, parallel-group comparison of 2 data extraction processes, as outlined in Appendix Figure 1 (available at Annals.org). For each review, the investigators organized 2 independent data extraction teams. Team 1 (human-only) consisted of investigators who extracted data from the original study report and others who reviewed and revised the extracted information for completeness and correctness against the original study report. Team 2 (AI-assisted) uploaded the full-text PDF of each study to the LLM Claude for initial data extraction; a human investigator then reviewed and revised the AI-generated output for completeness and correctness using the original study report. Both teams followed the same data extraction instructions provided by the principal investigator. The teams consisted of different individuals, all with prior but varying experience in data extraction (Part 1 of the Supplement, available at Annals.org) and a thorough understanding of the review's topic and objectives.

To ensure consistency and refine the process, both teams participated in an initial pilot data extraction using the same 2 pilot studies. Team members independently extracted data and compared results to assess consistency. On the basis of discrepancies, the principal investigator revised the data extraction instructions as needed. The pilot aimed to standardize procedures, enhance understanding of the data extraction tasks, and test the usability and clarity of the data extraction forms. Team 2 also used the pilot phase to draft and iteratively refine prompts for the LLM to ensure effective AI-assisted extraction. After the pilot phase, both teams independently extracted data from the same selected studies, without access to each other's results.

Blinded adjudicators uninvolved in any of the 6 review teams or the data extraction process (called the adjudication team in Appendix Figure 1) compared the results from the 2 data extraction teams. In cases of discrepancies between extraction teams, 2 adjudicators independently reviewed the original study reports to determine which extracted data were accurate. A third, senior adjudicator resolved any discrepancies and inconsistencies between the initial adjudicators.

Prompt Engineering and Output Generation

We chose Claude Pro, versions 2.1, 3.0 Opus, and 3.5 Sonnet (Anthropic), for data extraction (20). According to Anthropic's official policies, Claude Pro does not use uploaded PDFs or user interactions to train its models unless explicitly permitted, which respects the intellectual property rights of copyrighted study reports. Investigators received training on the use of Claude and the basics of prompt engineering. Prompt engineering involves designing text inputs (prompts) with the objective of accurate and succinct output from LLMs. Investigators conducting the reviews received a pool of example prompts, which were successfully used during a proof-of-concept study for data extraction with Claude 2.0 (6). The training emphasized

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that prompt iteration is often necessary to improve accuracy. During the pilot phase, initial prompts were developed on the basis of typical instructions that would be given to human extractors. Outputs were then evaluated for correctness and completeness, and prompts were refined accordingly. Throughout the study, investigators could consult a data scientist with expertise in prompt engineering as needed. Because this was a pragmatic study, each review team determined the extent of prompt refinement and the number of iterations required to achieve satisfactory results. When outputs were overly verbose, investigators edited them for brevity without altering the requested content. Part 2 of the **Supplement** presents the prompts used for each review.

Outcomes and Error Analysis

The primary outcomes of the study were the concordance of data extracted by the 2 approaches and the time required to complete the data extraction for each study report. Concordance was defined as "the proportion of data items that are factually congruent between the 2 methods, regardless of differences in style or presentation" and measured as the percentage agreement between approaches. To measure concordance, we used percentage agreement rather than κ statistics because chance agreement is minimal when extractors rely directly on source material. No universal percentage agreement thresholds exist for AI data extraction; most studies use pragmatic cutoffs (11).

The time required for data extraction included the total time spent on all tasks necessary for completion and verification, including prompt engineering during the pilot phase and any subsequent revisions made for individual studies in the AI-assisted process.

The secondary outcomes were the accuracy, sensitivity, positive predictive value, and F1 score of the AI-assisted approach (see **Appendix Figure 2** [available at Annals.org] for definitions) and the types of errors made by each of the data extraction approaches. In the context of data extraction, accuracy refers to the proportion of data items that were either correctly extracted or correctly identified as unavailable. To calculate these performance metrics, we used the finalized data by the adjudication team as the reference standard because that process can be considered an enhanced version of human data extraction.

Two adjudicators independently classified error types (for example, missed data, misallocated data, or fabricated data) and the potential effect of errors (inconsequential, minor, or major). A third adjudicator resolved any discrepancies in adjudicator classifications.

Sample Size Calculation and Data Analysis

Our proof-of-concept study found an 83% concordance between human data extraction and LLM-conducted data extraction (6). For the current study, we aimed to estimate the proportion of concordant data item pairs across 4 categories: study characteristics, participant characteristics, interventions and participant flow, and outcomes. To achieve a 2-sided 95% Clopper–Pearson CI that is approximately 7 percentage points wide, we required 500 data items for an observed binomial proportion of 83%.

To assess the time required for data extraction, investigators tracked their time while working on any of the tasks of the 2 data extraction processes. For team 2, this included development and refinement of the prompts.

The unit of analysis was the extracted data item as defined in each systematic review. To avoid disproportionate weighting of incorrectly extracted data items, we excluded data items that were subordinate to missed primary data items. For instance, if one of the extraction methods missed an entire outcome, it was recorded as an error. Subordinate elements of the missed outcome—such as the point estimate, CI, *P* value, or length of follow-up—were not counted as additional errors.

In line with sample size considerations, we present proportions along with 95% Clopper–Pearson CIs for concordance, accuracy, sensitivity, and positive predictive value, where the number of true positives is considered binomial with the sample size equal to the corresponding denominator.

For time on task and types and severities of errors, we conducted descriptive analyses. Because of the exploratory nature of this study, we made no adjustments for multiplicity in the analyses. This allowed us to explore the data comprehensively and generate valuable insights without imposing strict adjustments for multiple comparisons. In line with the study objectives outlined in the protocol, we deliberately refrained from conducting statistical tests to compare the human-only and AI-assisted approaches. These comparisons are intended to be exploratory rather than confirmatory.

Role of the Funding Source

The Agency for Healthcare Research and Quality and RTI International Innovation Office were not involved in the design, conduct, or data analysis of this study.

RESULTS

The 6 systematic reviews in the SWAR yielded 9341 data elements from 63 studies. Three reviews focused exclusively on randomized controlled trials, whereas the others included both randomized and nonrandomized studies (**Appendix Table**, available at Annals.org). **Supplement Table 1** (available at Annals.org) presents characteristics of the reviewers extracting data.

Concordance Between Data Extraction Approaches

The overall concordance between the 2 data extraction approaches was 77.2% (95% CI, 76.3% to

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Table 1. Performance Metrics of the AI-Assisted Approach for Each Review and the SWAR Overall

Study, Year (Reference)	Concordance (95% CI), %	Accuracy (95% CI), %	Sensitivity (95% CI), %	PPV (95% CI), %	TP, *n*	FP, *n*	FN, *n*	TN, *n*
AHRQ, 2024 (14)	65.5 (63.2–67.7)	88.2 (86.6–89.7)	86.3 (84.4–87.9)	99.8 (99.4–100)	1319	2	210	266
USPSTF, 2023 (15)	85.1 (83.8–86.3)	93.5 (92.6–94.3)	94.3 (93.5–95.1)	98.9 (98.6–99.1)	2908	38	175	144
Gartlehner et al, 2025 (18)	88.4 (86.2–90.4)	94.5 (92.8–95.9)	88.0 (84.5–91.0)	100 (99.0–100)	360	0	49	482
Ahluwalia et al, 2025 (19)	68.6 (66.0–71.1)	86.4 (84.4–88.2)	86.1 (84.0–88.0)	99.2 (98.4–99.6)	1071	9	173	84
AHRQ, 2024 (16)	76.4 (74.4–78.4)	92.2 (90.8–93.4)	84.1 (81.2–86.7)	96.3 (94.6–97.7)	607	23	115	1014
USPSTF, 2024 (17)	69.9 (64.2–75.1)	83.9 (79.2–87.9)	81.0 (75.6–85.7)	100 (98.2–100)	201	0	47	44
Combined across reviews	77.2 (76.3–78.0)	91.0 (90.4–91.6)	89.4 (88.6–90.1)	98.9 (98.6–99.1)	6466	72	769	2034

AHRQ = Agency for Healthcare Research and Quality; AI = artificial intelligence; FN = false negative; FP = false positive; PPV = positive predictive value; SWAR = study within reviews; TN = true negative; TP = true positive; USPSTF = U.S. Preventive Services Task Force.

78.0%) (Table 1). Concordance was highest for data describing study characteristics (83.3% [CI, 80.3% to 86.1%]) and lowest for data on outcomes and results (75.9% [CI, 74.8% to 76.9%]) (Appendix Figure 3, available at Annals.org). Levels of concordance varied from 65.5% to 88.4% across the 6 individual reviews (Table 1).

Of the 9341 extracted data item pairs, 2132 were discordant between human-only and AI-assisted extractions. Compared with the adjudicated reference standard, human-only extractions were incorrect in 41.7% ($n = 890$) of the discordant items, whereas AI-assisted extractions were incorrect in 32.9% ($n = 701$). In some instances, both approaches had incorrect extractions (6.6%; $n = 140$) compared with the adjudicated reference standard. In 18.8% ($n = 401$) of discordant pairs, neither extraction was wrong (Appendix Figure 4, available at Annals.org). Such situations arose when the reason for discordance was the level of reported detail, poor definition of data elements, or ambiguous prompts. For instance, one approach might have extracted intention-to-treat results while the other extracted per protocol results because of an unclear definition of the data item or a prompt that did not specify which results to extract.

Performance of Human-Only and AI-Assisted Data Extraction

Compared with the adjudicated reference standard, the human-only data extraction approach had an accuracy of 89.0% (CI, 88.3% to 89.6%) and the AI-assisted approach an accuracy of 91.0% (CI, 90.4% to 91.6%). Accuracy refers to the percentage of data items correctly identified, either as accurate extractions when data were available or as reported absences when data were missing. Sensitivity, which measures the percentage of accurately extracted data items out of all available data items in study reports, was 86.5% (CI, 85.7% to 87.3%) for the human-only approach and 89.4% (CI, 88.6% to 90.1%) for the AI-assisted approach. Positive predictive value, indicating the percentage of correctly extracted data items out of all extracted data items, was 99.2% (CI, 99.0% to 99.4%) for the human-only approach and 98.9% (CI, 98.6% to 99.1%) for the AI-assisted approach. Figure 1 shows these performance metrics graphically. The F1 scores, representing the harmonic means of sensitivity and positive predictive value, were 0.92 and 0.94, respectively.

Table 1 presents performance metrics of the AI-assisted approach for each included review.

Appendix Figure 3 (available at Annals.org) illustrates accuracy, sensitivity, and positive predictive value stratified by data category (participant characteristics, study characteristics, intervention and participant flow, and outcomes and results). Overall, performance metrics were consistent for different data categories. Supplement Table 2 (available at Annals.org) presents these results by individual review.

Error Analysis

Table 2 describes the different error types, the potential effect of errors, and the frequency of each error type out of all extracted items. Among the 9341 extracted data items, incorrect extractions were identified by the adjudicated reference standard in 1731 cases (AI-assisted, 9.0% [CI, 8.4% to 9.6%] vs. human-only, 11.0% [CI, 10.4% to 11.7%]) (Figure 1). The risk for major errors—those that significantly compromise data accuracy and could lead to erroneous conclusions if left uncorrected—was similar between the 2 groups (AI-assisted, 2.5% [CI, 2.2% to 2.8%] vs. human-only, 2.7% [CI, 2.4% to 3.1%]). The most common errors for both approaches were missed data (AI-assisted, 5.5% vs. human-only, 6.2%) and misallocated data (AI-assisted, 1.8% vs. human-only, 1.9%). Incorrectly inferred or fabricated data, which refers to data items that were not present in source study reports but were captured by human data extractors or were erroneously generated by the LLM (hallucinated) and not corrected by the human reviewer, were rare in both approaches (AI-assisted, 0.8% vs. human-only, 0.5%).

Time on Task

Across the 6 included reviews, the AI-assisted approach reduced the median time required for data extraction by 41 minutes per study compared with human-only extraction (84 vs. 125 minutes). The range of time differences across the 6 reviews spanned from 78 minutes shorter to 25 minutes longer (Figure 2). In 5 of the 6 reviews, data extraction with AI assistance required less time than human-only efforts.

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Additional Exploratory Analyses

In post hoc analyses, we investigated whether the basic data types of the extracted information influenced concordance and performance metrics. We categorized the data into 3 types: numerical, string, and mixed data. For the purpose of this study, numerical data referred to data in which numbers provided the primary information for extraction, even if characters were also included (for example, 100 participants). String data, on the other hand, involved information where characters, words, or sentences were most relevant, regardless of the presence of numerals (for example, follow-up with 12-lead electrocardiography). Mixed data consisted of both numerals and characters, where information from both was equally important (for example, drug dosage of 20 mg). Overall, string data were extracted with higher concordance and better accuracy and sensitivity than numerical or mixed data types (**Appendix Figure 5** and **Supplement Table 3**, available at Annals.org).

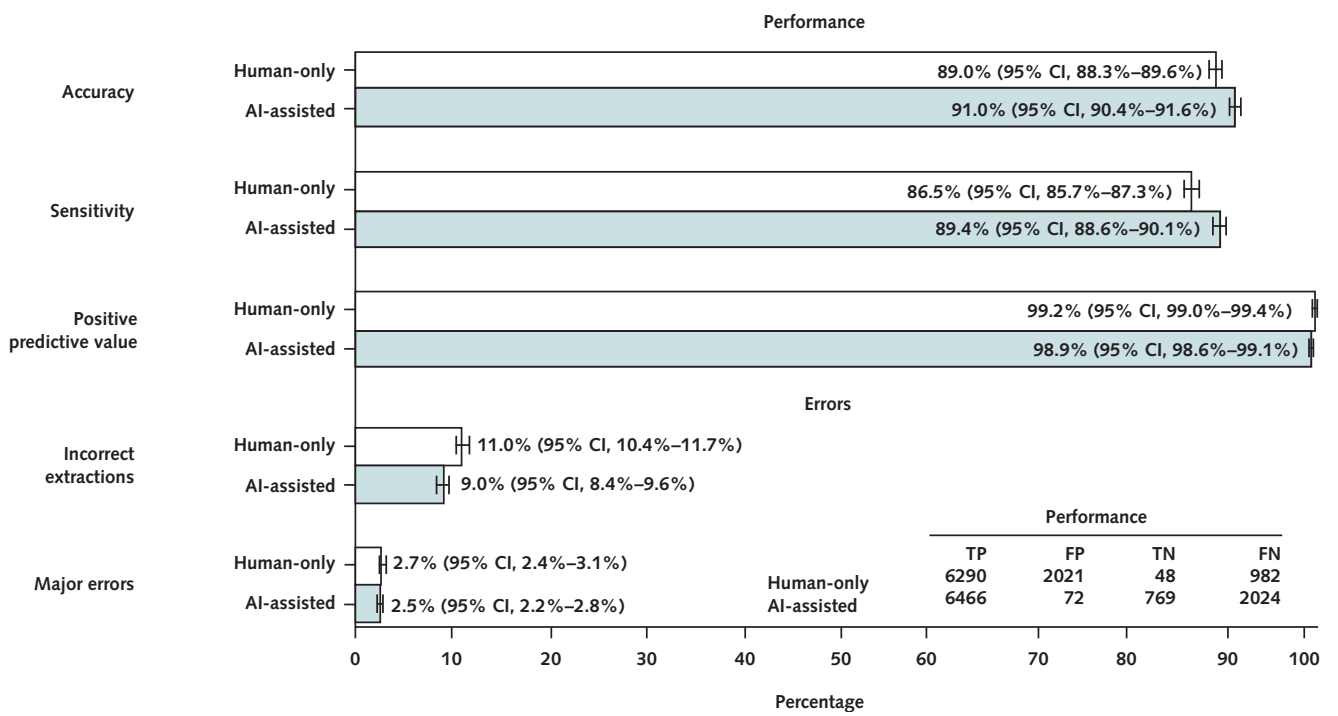
DISCUSSION

Our SWAR used 6 ongoing systematic reviews to prospectively evaluate the workflow of AI-assisted data extraction under real-world settings. Concordance between the 2 approaches was moderate, partly due to the error-prone nature of traditional human-only data extraction, as documented in previous studies (3, 21). Although all extractions were done by experienced reviewers and checked for correctness and completeness by a second investigator, 11% of the data items extracted by humans contained 1 or more errors. Most were minor and inconsequential, yet they contributed to discordant data extractions. The fallible nature of human data extraction was further confirmed by the AI-assisted teams' performance. Most errors in this group were missed data items that were overlooked by both the LLM during the initial extraction and the human investigators who reviewed the LLM's output.

Because there are currently no published workflow validation studies of AI-assisted data extraction, no direct comparison of findings with other studies is possible. Existing model validations of LLMs for data extraction reported accuracies ranging from 72% to 100%, consistent with the accuracy observed in our SWAR (6–9, 11).

Our study has several notable strengths. A primary strength is its design as a prospective workflow validation study. Compared with model validation studies that examine the performance and reliability of an LLM under specific, controlled conditions using already existing data sets, workflow validation studies integrate the use of the LLM into the workflow of an ongoing review, providing a detailed perspective on the model's practical effectiveness, efficiency, and utility (22). The prospective design protects against data contamination, which can arise when the data set used to evaluate performance also contributed to LLM training, potentially inflating its

Figure 1. Accuracy, sensitivity, positive predictive value, and errors of human-only and AI-assisted data extraction.



AI = artificial intelligence; FN = false negative; FP = false positive; TN = true negative; TP = true positive.

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Table 2. Types and Frequencies of Errors and Their Potential Effects

Type of Error	Definition	Errors (Percentage of All Extracted Data Items) ($n = 9341$), *n (%)*	
		Human-Only Approach*	AI-Assisted Approach*
Missed or omitted data	Data available in the source study report that were either missed or omitted by the data extraction process	578 (6.2)	516 (5.5)
Misallocated data	Data provided in the source study report that were allocated to the wrong data extraction field	181 (1.9)	170 (1.8)
Incorrectly inferred or fabricated data	Data not available in the source study report that were captured by human data extractors or erroneously generated (hallucinated) by the LLM and not corrected by the human reviewer	48 (0.5)	72 (0.8)
Difference in level of detail†	The level of detail captured is the only difference between data extractions	47 (0.5)	37 (0.4)
Incorrect calculations	Mathematically incorrect calculations of data elements based on information provided in the source study report, including rounding errors	20 (0.2)	15 (0.2)
Other	Optional "other" field if none of the above apply; for example, typos or obvious misunderstandings of instructions	156 (1.7)	31 (0.3)
Potential effect of error			
Major error	An error that significantly compromises the correctness of the data and, if uncorrected, could lead to erroneous conclusions; for example, grossly incorrect calculations, misallocated data that result in a different interpretation, or fabricated data that result in a new or different interpretation of the data	255 (2.7)	231 (2.5)
Minor error	An error that is less severe than a major error and may or may not affect interpretation of the existing data; for example, small calculation errors or rounding errors that do not critically affect the data's overall utility, additional or alternative language describing study inclusion/exclusion criteria or the intervention that do not alter the meaning in a significant way	683 (7.3)	496 (5.3)
Inconsequential difference	A difference that most likely would not affect the interpretation of the data; for example, in the level of detail of inclusion criteria	92 (1.0)	114 (1.2)

AI = artificial intelligence; LLM = large language model.
* When both approaches produced incorrect extractions for the same data item, errors were attributed to each approach independently. As a result, the total number of recorded errors exceeds the total number of incorrectly extracted data items.
† Not an error per se but a discrepancy that influenced concordance.

capabilities (23). Because developers of LLMs rarely disclose their training data sets, the prospective design of our study using ongoing, unpublished reviews provides a rigorous safeguard against this risk (22).

In addition, to protect against benchmark bias, a form of classification bias that can arise when the reference standard is imperfect, we did not consider the human-only extraction as the reference standard. Instead, we verified each discrepancy between human-only and AI-assisted extractions against the original study report by blinded adjudicators. This allowed a more accurate assessment of the performance of both approaches.

Another notable strength is the inclusion of a wide range of systematic review topics and study designs, encompassing both randomized controlled trials and nonrandomized studies of interventions. This diversity enhances the study's generalizability across research contexts, especially given that performance metrics showed no substantial variation across the individual reviews. Finally, the large data set, comprising 9341 data item pairs, ensures sufficient statistical power for robust analyses.

Despite the methodological strengths, our study also has several limitations. As with any workflow validation study, the generalizability of the findings to other types of data and evidence synthesis may be restricted. Although the 6 included reviews covered a broad range of topics, all were commissioned by the Agency for Healthcare Research and Quality and focused on health care interventions. Results could differ for reviews of diagnostic test accuracy, prognostic studies, or studies in other sectors (such as education, economics, or business). Another limitation stems from human variation. Differences in systematic review experience, levels of detail in data extraction, team approaches to verifying data extractions, and proficiency in engineering prompts may influence the reproducibility and the broader applicability of our results.

Although time spent on task was one of our primary outcomes, consistently tracking time across reviews was challenging. For example, rate restrictions of the LLM due to factors like institutional traffic varied across institutions and limited the workflow at 1 institution. We assumed that time tracking for each review was done under similar conditions, so the absolute differences in data extraction time accurately reflect the results within each review. However, comparisons of time spent across different reviews may be less reliable. Nonetheless, the consistent time savings observed in the AI-assisted

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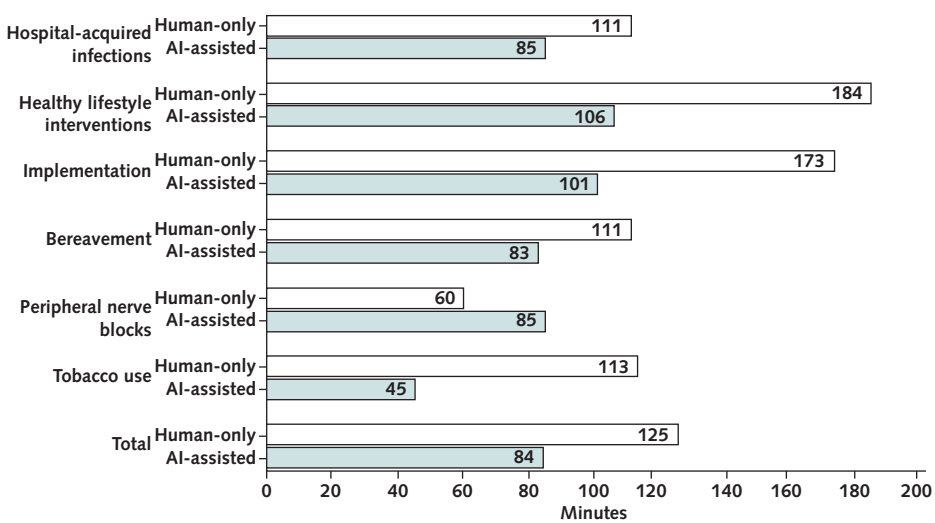
approach in 5 of 6 reviews strengthen confidence in the overall direction of the results. In the future, AI-assisted data extraction will likely be integrated into systematic review software, streamlining the process and making it more time-efficient than in our study. However, it will always be essential to develop high-quality data extraction forms with clear definitions and instructions.

Assessing the concordance of data extractions and classifying errors often required subjective judgment. To limit subjectivity, 2 study investigators from the adjudication team independently verified errors and classified their type and severity and a third investigator resolved any discrepancies. Nevertheless, inconsistencies in classifications are likely, and different investigators may have reached different conclusions about the types and severities of errors. Although we attempted to blind the adjudication team to the data extraction approach, in some cases it was apparent which extraction was from the AI-assisted approach due to the verbose nature of the responses, which sometimes included irrelevant information. Furthermore, because we adjudicated only discordant data extractions, we have missed instances where extraction results were concordant but both extractions were incorrect. However, we assume that such cases are rare. To avoid disproportionate weighting of incorrectly extracted data items, we excluded their subordinate elements from our analyses. Including them would alter the overall error proportions in both groups, although the relative difference between approaches would likely remain similar. Because of the rapid advancement of LLMs, we switched from Claude 2 to Claude 3 during the SWAR. This change was in line with our protocol but may have contributed to differences in performance metrics across reviews. Since the completion of our study, Claude 4 has been released. It remains unclear whether this newer version would yield improved performance, or how data extractions using Claude compare with those using other LLMs (such as ChatGPT and Gemini).

Finally, although our adjudicators assessed the *potential* effect of errors, we did not assess the *actual* effect of data extraction errors on systematic review results and conclusions. The consequences of inaccuracies in different data elements can vary, potentially affecting a review's conclusions differently. In our study, the adjudicators who judged the seriousness of the error were experienced systematic reviewers but were not part of the teams for each systematic review.

Although the results of our study are encouraging, several key areas warrant further investigation to advance the field. Our research focused on a single scenario where the LLM did the initial data extraction and human investigators verified its work. However, many alternative scenarios could be explored. For instance, overall performance might improve if the roles were reversed, with an LLM verifying human data extraction, or if a second LLM reviewed the work of the first LLM before human verification. Ultimately, a prospective workflow validation of extraction by LLMs alone without human review would help identify where humans must stay involved. Any advancement in the application of LLMs to the data extraction process also needs to be validated. In this study, investigators manually developed and entered prompts into the LLM's web-based user interface. Developers of systematic review software are already working to integrate LLMs directly into software applications. Such integration could provide greater time savings and improved performance but might reduce flexibility if users cannot adapt prompts to their specific review

Figure 2. Median time spent for data extraction per study.



AI = artificial intelligence.

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needs. Developing standardized prompts for common data items could enhance LLM performance and ensure consistency across various contexts. In-depth case studies comparing LLM-assisted and human-only methods should extend beyond assessing the correctness of data extraction. They should also examine how inaccuracies affect evidence synthesis, such as meta-analyses, and influence the ultimate conclusions of systematic reviews.

Future research also needs to develop methods to assess the extent to which data contamination can bias LLM evaluations. For example, developing a private benchmark data set (24) (that is, where each data item is annotated with specific labels and the data set remains inaccessible to LLM models during training) based on full-text study reports from systematic reviews could provide a standardized reference for accuracy assessments. Such a data set would also eliminate reliance on human data extraction as a reference standard.

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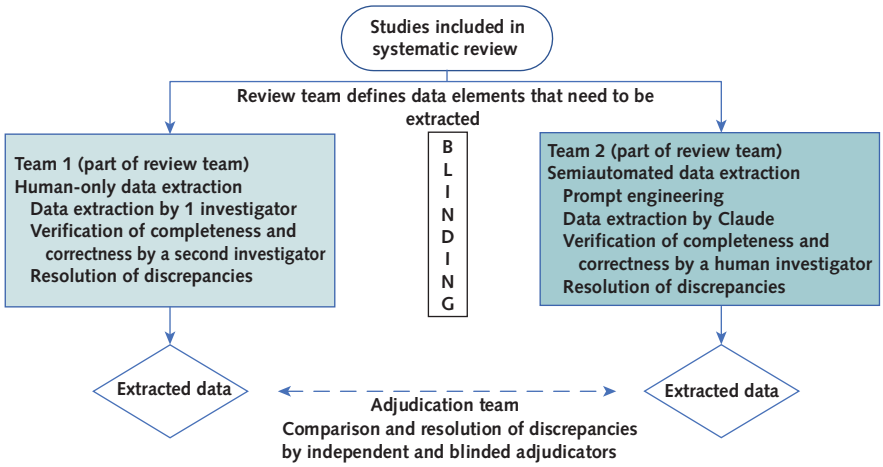
Appendix Table. Topics, Eligible Study Designs, and Numbers of Contributing Studies and Data Items of the Contributing Systematic Reviews

Study, Year (Reference)	Number of Included Studies (Eligible Study Designs)	Extracted Data Items, *n*	Version of Claude Used
AHRQ, 2024 (15)	10 (RCTs, NRSIs)	1797	3.5 Sonnet
USPSTF, 2023 (16)	7* (RCTs)	3265	3.5 Sonnet
Gartlehner et al, 2025 (19)	11 (RCTs, NRSIs)	891	2.1 and 3.0 Opus
Ahluwalia et al, 2025 (20)	20* (RCTs, NRSIs)	1337	2.1 and 3.0 Opus
AHRQ, 2024 (17)	8 (RCTs)	1759	2.1
USPSTF, 2024 (18)	7 (RCTs)	292	3.5 Sonnet
Combined across reviews	63	9341	-

AHRQ = Agency for Healthcare Research and Quality; NRSI = nonrandomized study of intervention; RCT = randomized controlled trial; USPSTF = U.S. Preventive Services Task Force.
* Studies were randomly sampled from a larger number of included studies.

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Appendix Figure 1. Outline of study design comparing 2 data extraction processes.



Appendix Figure 2. Definitions of performance metrics.

FN: The number of data items missed or incorrectly extracted from the study report.

FP: The number of data items for which data were extracted when no data were available in the study report (i.e., incorrectly inferred or fabricated data).

TN: The number of data items that were correctly identified as not available in the study report.

TP: The number of data items correctly extracted from the study report.

Accuracy: The proportion of data items that were either correctly extracted from the study report or correctly noted as missing:

$$\frac{(TP + TN)}{(TP + FP + TN + FN)}$$

Positive predictive value (= precision): Of all extracted data items, the proportion that were correctly extracted:

$$\frac{TP}{(TP + FP)}$$

Sensitivity (= recall): Of all data items reported in a study report, the proportion that were correctly extracted:

$$\frac{TP}{(TP + FN)}$$

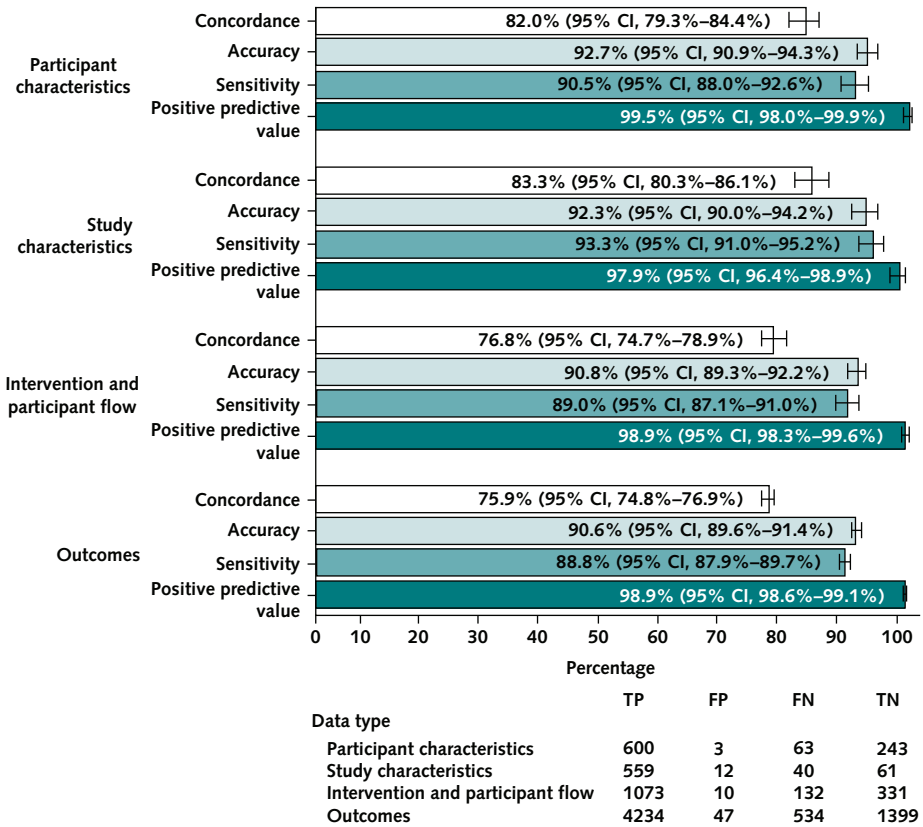
F1 score: An evaluation metric that combines the positive predictive value and sensitivity (via harmonic mean) into a single proportion that ranges between 0 (poor) and 1 (perfect). It balances both FP and FN, making it especially useful for imbalanced data sets:

$$2 * \frac{positive\ predictive\ value * sensitivity}{(positive\ predictive\ value + sensitivity)}$$

FN = false negative; FP = false positive; TN = true negative; TP = true positive.

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Appendix Figure 3. Concordance, accuracy, sensitivity, and positive predictive value for AI-assisted data extraction, stratified by data category.

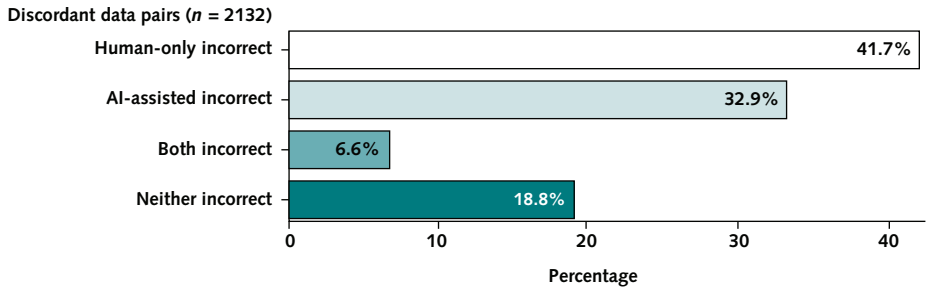


Data type	TP	FP	FN	TN
Participant characteristics	600	3	63	243
Study characteristics	559	12	40	61
Intervention and participant flow	1073	10	132	331
Outcomes	4234	47	534	1399

AI = artificial intelligence; FN = false negative; FP = false positive; TN = true negative; TP = true positive.

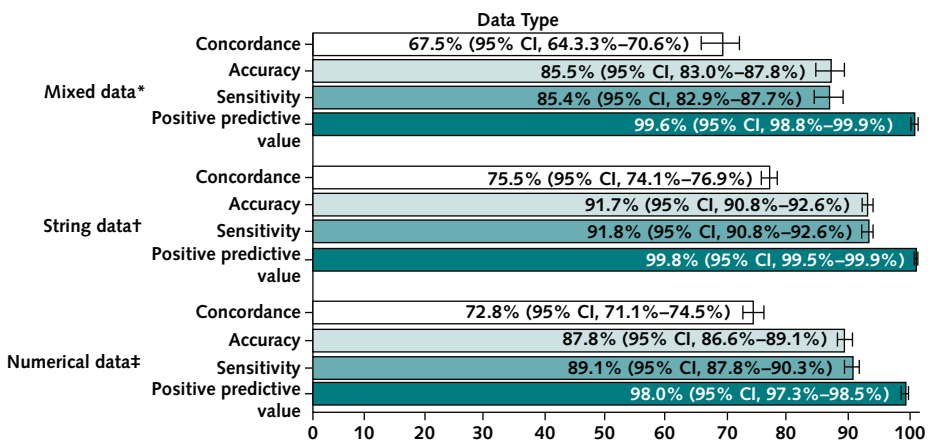
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Appendix Figure 4. Sources of error in discordant data items between human-only and AI-assisted approaches.



The analysis is limited to discordant data items ($n = 2132$). AI = artificial intelligence.

Appendix Figure 5. Concordance, accuracy, sensitivity, and positive predictive value for AI-assisted data extraction, stratified by basic data type.



Data type	TP	FP	FN	TN
Mixed data	737	3	126	24
String data	3282	8	296	8
Numerical data	2284	47	280	80

The analysis is limited to data items with available data ($n = 7252$). In 2089 cases, no data were reported; these cases contributed to the overall analyses but not to the analyses stratified by basic data type. AI = artificial intelligence; FN = false negative; FP = false positive; TN = true negative; TP = true positive.

* Consists of both numerals and characters, where information from both was equally important (e.g., drug dosage of 20 mg).

† Involves information where characters, words, or sentences were most relevant, regardless of the presence of numerals (e.g., follow-up with 12-lead electrocardiography).

‡ Refers to data in which numbers provided the primary information for extraction, even if characters were also included (e.g., 100 participants).

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